

Driver's Facial Expression Recognition using Global Context Vision Transformer

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Abstract—Driver's facial expression recognition plays a critical role in enhancing driver safety, comfort, and overall driving experience by proactively mitigating potential road risks. While most existing works in this domain rely on CNN-based approaches, this paper proposes a novel method for driver facial expression recognition using Global Context Vision Transformer (DFER-GCViT). With its inherent capabilities of transformer-based architectures and global context modeling, the proposed method handles challenges commonly encountered in real-world driving scenarios, including occlusions, head pose variations, and illumination conditions. Our method consists of three modules: preprocessing for face detection and data augmentation, facial feature extraction of local and global features, and expression classification using a modified GC-ViT classifier. To evaluate the performance of DFER-GCViT, extensive experiments are conducted on two benchmarking datasets namely the KMUFED driver facial expression dataset and FER2013 general facial expression dataset. The experimental results demonstrate the superiority of DFER-GCViT in accurately recognizing driver's facial expressions, achieving an average accuracy of 98.27% on the KMUFED dataset and 73.78% on the FER2013 dataset, outperforming several state-of-the-art methods on these two benchmarking datasets.

Index Terms—driver's facial Expression recognition, driver emotion recognition, vision transformer, global context vision transformer

I. INTRODUCTION

With the rapid advancements in technology, a variety of techniques have emerged to improve road safety and mitigate the risks associated with human factors during driving. One critical aspect in this domain is the recognition of the driver's facial expressions, which plays a crucial role in enabling monitoring systems such as the Advanced Driver Assistance System (ADAS) to understand the driver's emotional state and distraction. This understanding allows ADAS to adapt its responses, providing appropriate assistance or alerts when necessary. Extensive research has established a strong correlation between facial expressions and emotions [1] [2]. This knowledge has led to the development of facial expression recognition systems capable of effectively interpreting and classifying various emotional states, including the six basic emotions defined by Ekman *et al.* [3]: anger, disgust, fear, happiness, sadness, and surprise. By accurately capturing and interpreting the facial expressions of drivers, systems can gain profound insights into the driver's emotional driving process, thus enabling enhanced driver safety, comfort, and overall driving experience.



Fig. 1. Facial expression recognition in driving environments may involve additional challenges such as changing lighting conditions, occlusions, pose variations, and motion blur [4].

However, despite the significant progress in facial expression recognition generally [5], most of the proposed models seem to mainly work well under controlled environments (eg. shopping malls) but suffer in dealing with real-world scenarios such as the driving environment (See Fig. 1) which involves additional challenges seen in changing lighting conditions, occlusions, and pose variations.

Previous works in the literature of Driver Facial Expression Recognition (DFER) have adopted different machine learning (ML) and deep learning (DL) approaches e.g. [6], [4]. Most of the proposed methods are mainly based on variants of traditional Convolutional Neural Networks (CNNs), with relative success while sometimes struggling in capturing the fine-grained specificities of the driver environments. In this context, we propose in this paper a novel transfer learning approach called DFER-GCViT for more accurate driver facial expression recognition. Our approach is based on Vision Transformers (ViT) using the Global Context (GC-ViT), which particularly addresses the limitations of the ViT architectures seen in (1) the loss of local information which is crucial for accurate expression recognition, (2) the lack of inductive bias that enhances the model's ability to recognize expressions by enhancing the features, and (3) high computational cost which may make the system less practical for real-time implementations.

The main contributions of this work can be summarized as follows:

- We investigate the use of Global Context Vision Transformers for facial expression recognition, with a specific focus on the driver environment. To the best of our knowledge, this is the first study that explores the potential of GC-ViT in this domain.

- We propose a novel approach called DFER-GCViT, which is based on transfer learning and extends the classifier head of the GC-ViT architecture. Our approach aims to enhance the model's performance in accurately recognizing driver facial expressions.
- We tackle the limitations faced by ViT models in the context of facial expression recognition, as well as the challenges posed by real-world scenarios through the inherent advantages of GC-ViT architecture.
- We conduct comprehensive comparisons with state-of-the-art methods on two datasets: the Keimyung University Facial Expression of Drivers (KMU-FED) dataset and the Facial Expression Recognition 2013 (FER2013) dataset.

II. RELATED WORK

In this section, we discuss the recent advances in the field of recognizing driver facial expressions, emphasizing their significance in improving driving performance and road safety. Researchers have indeed proposed various techniques to accurately identify and classify facial expressions or emotions displayed by drivers.

Previous works on driver emotion recognition have explored different signals. For example, in [7], a convolutional neural network (CNN) based classifier was implemented to accurately classify three physiological signals, including Electrodermal Activity (EDA), Skin Temperature, and the Electrocardiogram (ECG) into four emotional states. Other researchers (e.g. [8] [9]) focused on utilizing Electroencephalography (EEG) signals to detect the emotional states of drivers. Furthermore, facial expressions have gained prominence as a significant cue in emotion recognition, leading to the development of various methods for driver facial expression recognition. These methods can be broadly categorized into traditional machine learning-based (ML), deep learning-based (DL), and hybrid approaches. Traditional ML-based approaches, such as the lightweight multilayer random forest (LMRF) model proposed in [10], achieved high accuracy of 93.4%, 77.4%, and 95.1% on CK+, MMI, and KMU-FED datasets respectively, using features extracted based on distance ratios and angles. Another study in [6] presented a novel method that fused local binary patterns (LBP) and facial landmark features, achieving notable accuracy of 86.7% on the CK+ dataset using a Support Vector Machine (SVM) for classification.

DL-based approaches, particularly CNNs, have significantly advanced the field of driver facial expression recognition, and have showcased outstanding performance across diverse datasets. For instance, Jain *et al.* [11] introduced the Squirrel Search Optimization with Deep Learning Enabled Facial Emotion Recognition (SSO-DLFER) system, which employed Neural Architectural Search (NASNet)-Large to extract features and a gated recurrent unit (GRU) classifier to classify emotions. In another study [12], a Very Deep Convolutional Networks (VGGNet) classifier was utilized to handle semi-facial occlusions during the recognition of driver facial expressions, with the calculation of optical flow. This approach achieved high accuracy of 95.92% on the CK+ and 94.78% on

KMU-FED datasets. Additionally, Zaman *et al.* [13] proposed an improved Faster R-CNN face detector coupled with a NasNet Large CNN classifier, achieving impressive accuracies on JAFEE, CK+, FER-2013, and AffectNet datasets. Sahoo *et al.* [14] introduced two customized CNN models comprising 6 and 10 layers, respectively. By employing transfer learning with a pre-trained VGG16 model, they achieved significant improvements - the 10-layer CNN model achieved the highest accuracy of 68.34% on the FER2013 dataset. Pant *et al.* [15] proposed a Convolutional Neural Network (CNN), such as ResNet, which achieved significant accuracy of 83% on the FER2013 dataset. Multi EmoNet methods [16] were also introduced to predict emotion under changes in illumination, utilizing CNN-based feature extraction with a simplified Xception model classifier, achieving an accuracy of 83.9% and 99.97% on the FER+ and CK+ datasets.

Hybrid methods combining ML, DL, and handcrafted features have also been explored. For instance, Sukhavasi *et al.* [17] proposed the use of Gabor and LBP feature extraction, followed by fusion and classification using SVM combined with CNN, achieving notable accuracy of 84.41%, 95.05%, 98.57%, and 98.64% on the FER2013, CK+, KDEF, and KMU-FED datasets, respectively.

Despite the promising performance achieved by the aforementioned methods, CNNs have some inherent limitations in modeling global context information in facial images, which is crucial for accurate expression recognition. Moreover, due to their limited receptive fields, CNNs tend to struggle to distinguish between subtle changes in facial expressions and to capture the long-range spatial dependencies between facial features. To partially overcome these issues, the Vision Transformer (ViT) architectures [17] has shown promising results in various computer vision tasks compared to the CNN methods. However, there is relatively limited research exploring the use of ViT in the field of facial expression recognition (FER). Kim *et al.* [18] has proposed an approach that incorporates ViT, such as Squeeze ViT and focus to capture both global and local features. Liu *et al.* [19] introduced the Patch Attention Convolutional Vision Transformer to address the occlusion problem. Xue *et al.* [20] proposed the TransFER model, which utilizes ViT to model the relationships between local regions. In the context of driver facial expression recognition using ViT, only two works have been reported so far. The first work [21] proposes using the original ViT architecture and a parallel imaging framework with the StarGAN network to enhance the model's performance. The second work [22] introduces a multi-modal Vision Transformer (ViT-DD) to detect driver distraction, utilizing a semi-supervised learning approach.

Motivated by the observations above and taking into consideration the limitations and advantages of ViT, we propose in the work a new ViT-based architecture for the task of driver facial expression recognition. We introduce Global Context Vision Transformers with transfer learning to avoid overfitting with insufficient training data, which is particularly important

for ViT models as they typically require large-scale datasets for effective training.

III. PROPOSED METHODOLOGY

The proposed methodology for driver facial expression recognition using GC-ViT comprises three main modules. Firstly, a pre-processing step is performed, which involves face detection using Multi-task Cascaded Convolutional Networks (MTCNN) [23]. The detected faces are then cropped, and data augmentation is applied to enhance the dataset. Secondly, the preprocessed images are fed into the DFER-GCViT model, where features are extracted and facial expressions are classified. Fig. 2 illustrates the architecture of our proposed model. The step-by-step process of the DFER-GCViT methodology is summarized in Algorithm 1.

Algorithm 1 Driver Facial Expression Algorithm using Transfer Learning DFER-GCViT

- 1: Split the dataset into training and testing sets.
 - 2: Apply appropriate pre-processing techniques to the dataset.
 - 3: Build the model architecture using the pre-trained GC-ViT network parameters.
 - 4: Fine-tune the GC-ViT model on the training data.
 - 5: Extract expressive features using the pre-trained GC-ViT model.
 - 6: Classify the expressions using the modified classifier based on the extracted features.
 - 7: Evaluate the performance of the DFER-GCViT model on the testing set.
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Before discussing the modules in detail, it is important to provide a clear definition of GC-ViT. GC-ViT is an architecture originally introduced by [24] in 2022, that incorporates several key components working together to achieve effective feature extraction and classification. It can be considered as a hierarchical ViT framework similar to Swin Transformer [25] but with unique blocks including a global query generator, a global self-attention module, and a downsampling operator (as shown in Fig. 3), differentiating it from Swin Transformer and the original ViT. The GC-ViT architecture begins with the Stem/PatchEmbed layer, responsible for dividing the input images of size $H \times W \times 3$ into patches and converting them into embeddings. These patches are then processed through the GC-ViT stages or levels, which comprise alternating local and global self-attention blocks. These blocks leverage a global query tokens generator to extract features. The calculation of the global query tokens can be expressed as follows:

$$X^i = F - \text{MBConv}(X^{i-1}) \quad (1)$$

$$X^i = \text{MaxPool}(X^i) \quad (2)$$

Here, X represents the feature maps, and i denotes the stage ($i \in \{1, 2, 3, 4\}$). $F\text{-MBConv}$ block refers to the Fused-MBConv block, while MaxPool represents the max pooling layer in the generator. the Fused-MBConv block is similar to

the one used in EfficientNetV2 [26] with certain extensions as below:

$$\begin{aligned} \hat{X} &= \text{DW-Conv}_{3 \times 3}(X), \\ \hat{X} &= \text{GELU}(\hat{X}), \\ \hat{X} &= \text{SE}(\hat{X}), \\ X &= \text{Conv}_{1 \times 1}(\hat{X}) + X \end{aligned} \quad (3)$$

SE , GELU , and $\text{DW-Conv}_{3 \times 3}$ represent the Squeeze and Excitation block, the Gaussian Error Linear Unit, and the 3×3 depth-wise convolution, respectively.

Furthermore, the global self-attention module utilizes the extracted global query tokens to interact with the key and value local features, unlike local attention, which focuses on learning key, value, and query local features. The computation of the global self-attention query, key, and value features can be described as follows:

$$\mathbf{Q}_g \in \mathbb{R}^{B^* \times C \times h \times w} := [\mathbf{q}_g, \dots, \mathbf{q}_g], \mathbf{q}_g \in \mathbb{R}^{B \times C \times h \times w} \quad (4)$$

$$\mathbf{q}_g \in \mathbb{R}^{B^* \times N \times C} \xleftarrow{\text{reshape}} \mathbf{Q}_g \in \mathbb{R}^{B^* \times C \times h \times w} \quad (5)$$

$$\mathbf{k}, \mathbf{v} = \mathbf{g}(\mathbf{x}) \in \mathbb{R}^{B^* \times N \times C} \quad (6)$$

q_g represents the global attention query with dimensions $B \times C \times h \times w$, where B, C, h , and w denote batch size, embedding dimension, local window height, and width, respectively. $B^* = B \times N$, where N represents the number of local windows. Moreover, the self-attention module is calculated as follows:

$$\text{Attention}(q_g, k, v) = \text{Softmax}\left(\frac{q_g k}{\sqrt{d}} + b\right) v \quad (7)$$

Here, the scaling factor is denoted as d and the learnable relative position bias term is represented by b .

A. Pre-processing Module

Firstly, we apply face detection using the MTCNN face detector to detect and localize the faces in the images. The detected faces are then cropped to focus on the facial region of interest. To increase the diversity and amount of training data, we apply various data augmentation techniques as geometric transformations. These techniques include Random Horizontal Flip, Random Rotation, Random Affine, and Random Perspective. These transformations introduce variations in facial orientation, pose, and perspective, helping the model to generalize better to different facial expressions and head orientations. Additionally, we filter the Gaussian blur to reduce noise and smooth out any imperfections in the data. This helps improving the overall quality of the images and enhancing the model's ability to capture meaningful facial features. After these augmentations, we resize the images to a standard input dimension of 224×224 pixels. This ensures consistency in the input size for our DFER-GCViT model. Furthermore, we standardize the pixel values across the images to help improving the convergence of the model during training and ensuring that the features have a consistent scale and range.

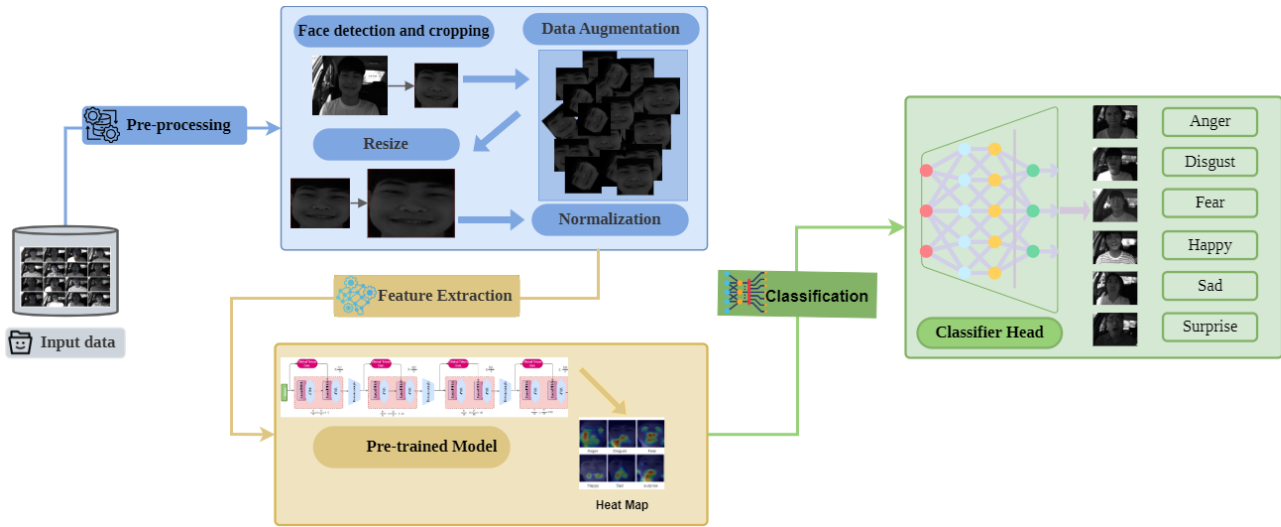


Fig. 2. The General workflow of our proposed methodology.

B. Feature Extraction Module

The GC-ViT architecture comprises two primary sections: feature extraction and classification. In this module, we focus on leveraging the feature extraction section to extract informative features from a new dataset. To accomplish this, we utilize the pre-trained weights and architecture of GC-ViT, allowing us to extract patterns and representations from preprocessed images with dimensions of $224 \times 224 \times 3$. Through this process, the images are transformed into a set of feature maps that encode hierarchical representations, capturing both local and global contextual information pertaining to facial expressions at each stage of the GC-ViT. The extracted features provide a rich representation of the input images, enabling subsequent classification tasks. The overall workflow is depicted in Fig. 3, providing a visual representation of the feature extraction process.

C. Driver Facial Expression Recognition Module

The driver facial expression recognition or classification module using GC-ViT is a crucial component responsible for identifying and categorizing facial expressions exhibited by drivers. In our approach, we have made a modification to the ClassifierHead component to enhance its performance. Instead of using a traditional (fc) layer, we have replaced it with a Sequential module that comprises two linear layers, a batch normalization layer, a ReLU activation layer, and a dropout layer. This modification, depicted in the ClassifierHead part of Fig. 3, is aimed at improving the model's ability to capture expressive features and adapt to the specific requirements of the dataset, such as the number of classes. The first linear layer (FC1) performs a linear transformation of the input, which allows the model to learn more expressive features from the data. The subsequent batch normalization layer normalizes the activations, leading to improved stability and faster convergence during training. The introduction of the ReLU activation function introduces non-linearity, allowing the model to

capture complex feature relationships and enhance its discriminative power. Additionally, the dropout layer is applied during training to mitigate overfitting by randomly dropping out a fraction of the activations, promoting better generalization to unseen data. The output of the second linear layer (FC2) serves as the final prediction for the facial expression recognition task, providing the corresponding classification results. This modified configuration of the ClassifierHead not only enhances the discrimination and generalization capabilities of the GC-ViT model but also allows for fine-tuning and adaptation to the specific characteristics of the facial expression recognition task. By leveraging the benefits of the Sequential module and the various layers within it, the model is expected to capture more expressive features and improve the performance.

IV. EXPERIMENTAL ANALYSIS

We evaluated the performance of our method on two publicly available benchmark datasets, namely KMU-FED (Keimyung University Facial Expression of Drivers) [27] and FER 2013 (Facial Expression Recognition 2013) [28]. In this section, we provide a detailed description of these datasets, along with the experimental setup, and a comparative analysis of the obtained results. Fig. 4 showcases some sample images from both datasets, providing visual illustrations of the data used in our evaluation.

A. Dataset Description

a) *KMU-FED Dataset*: The KMU-FED dataset [27] consists of real-time images captured during driving using Near-Infrared (NIR) cameras mounted on the dashboard or steering wheel. The dataset specifically focuses on capturing various facial expressions exhibited by drivers, including anger, disgust, fear, happiness, sadness, and surprise. It comprises a total of 55 image sequences, amounting to 1106 images, obtained from 12 subjects. The images in the dataset have a resolution of 1600×1200 pixels. The KMU-FED dataset serves as

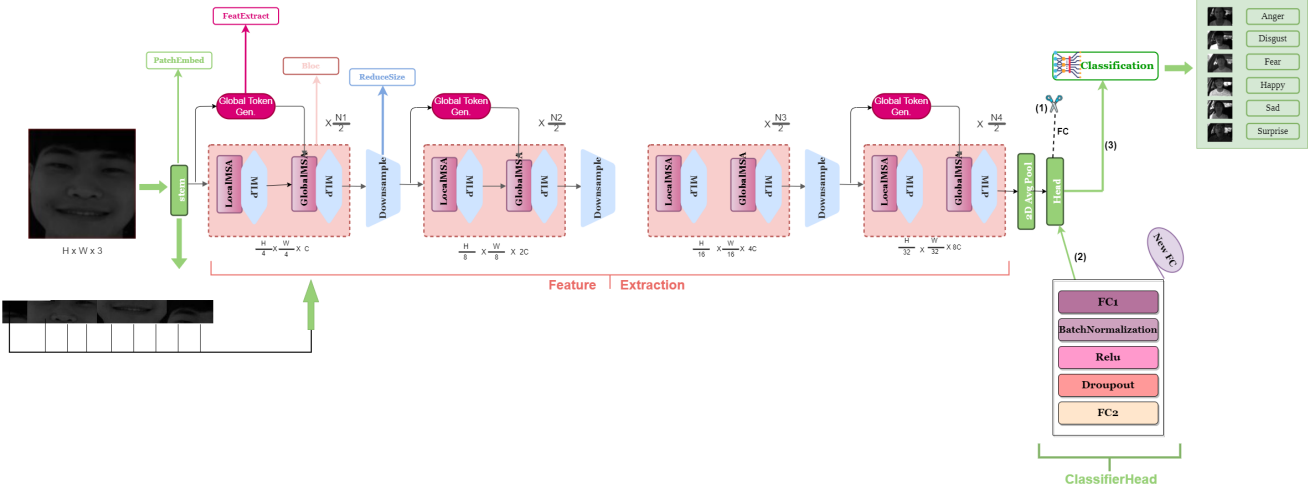


Fig. 3. The feature extraction and classification module's details with enhanced classifier head for driver facial expression recognition in GC-ViT, showing removing the fully connected layer (fc), adding a new fully connected layer (fc) to GC-ViT, and driver facial expression classification.



Fig. 4. Examples of images from (a) KMU-FED dataset, (b) FER2013 dataset.

a valuable resource for evaluating driver emotion detection algorithms in real-world driving scenarios. It provides an opportunity to study the impact of illumination variations and partial occlusions on facial expression recognition. To ensure an extensive analysis and fair comparison, we employed 10-fold cross-validation for evaluating the KMU-FED dataset.

b) FER 2013 Dataset: The FER2013 dataset [28] consists of 35,887 grayscale images of faces with a resolution of 48×48 pixels. The dataset is partitioned into three subsets: a training set containing 28,709 images, a public test set consisting of 3,589 images, and a private test set also comprising 3,589 images. The images in the dataset have undergone preprocessing to ensure that the faces are approximately centered and occupy a consistent amount of space across all samples. It covers a wide range of facial expressions, including anger, disgust, fear, happiness, sadness, surprise, and neutral. It serves as a standard benchmark for evaluating facial expression recognition algorithms and facilitates the assessment of algorithms' performance in capturing diverse facial expressions in unconstrained settings.

B. Implementation Setup

The proposed methodology, DFER-GCViT, was implemented using the open-source PyTorch framework on an NVIDIA GPU device, specifically the Quadro RTX 5000 with 16 GB of RAM. For the KMU-FED dataset, the training process utilized a batch size of 16. The Adaptive Moment Estimation (Adam) optimizer was employed with a fixed learning rate of 0.0001. The cross-entropy loss function was utilized, the model was trained for 60 epochs, and a 10-fold cross-validation scheme was applied. For the FER2013 dataset, a batch size of 16 was used. The Stochastic Gradient Descent (SGD) optimizer was employed with a momentum value of 0.9 and a learning rate decay of 0.002. Similar to the KMU-FED dataset, the cross-entropy loss function was used, and the model was trained for a total of 130 epochs. A summary of these implementation details is provided in Table I.

TABLE I
IMPLEMENTATION DETAILS AND HYPERPARAMETER SETTINGS FOR DFER-GCViT ON THE KMU-FED AND FER2013 DATASETS

Hyperparameters	KMU-FED dataset	FER2013 dataset
Optimizer	Adam	SGD
Loss Function	Cross-Entropy	Cross-Entropy
Batch size	16	16
Learning rate	0.0001	0.002
Epochs	60	130
Folds	10	/

C. Results and Discussion

To evaluate the effectiveness of our proposed DFER-GCViT method, we conducted comparative experiments on the KMU-FED and FER2013 datasets, comparing their performance against state-of-the-art methods.

a) Results on KMU-FED dataset: On the KMU-FED dataset, we conducted a comparative analysis between our

proposed DFER-GCViT model and three other methods to demonstrate the effectiveness of our modified version in improving accuracy. The three methods we compared with were as follows: (1) a hierarchical weighted random forest (hierarchical WRF) classifier that used geometric features [27], (2) a connected convolutional neural network (CCNN) that combines low-level and high-level features [29], and (3) a modified version of SqueezeNet 1.1, where the last layer has been adjusted [30]. Our proposed DFER-GCViT model achieved outstanding results with an average accuracy of 98.2%, surpassing all the compared state-of-the-art methods. The hierarchical WRF achieved an accuracy of 94.7%, resulting in a 3.5% difference compared to our DFER-GCViT model. The CCNN and modified SqueezeNet 1.1 achieved accuracies of 97.3% and 83.43% respectively, with differences of 0.9% and 14.7% compared to our proposed approach. It is worth noting that the modified SqueezeNet 1.1 showed a significant difference in performance. Table II summarizes these findings. Furthermore, we compared our proposed model with the original GCViT method without any modifications to the classifier head, but with the number of classes finetuned to match the number of classes in the dataset. We obtained an average accuracy of 97.6%, indicating that the proposed model with the new fc layer outperforms the model without any modifications in the fc layer, with a difference of 0.6%. This provides strong evidence for the effectiveness of the new fc layer in enhancing the performance of our proposed DFER-GCViT model.

TABLE II
COMPARISON BETWEEN THE PROPOSED METHOD AND STATE-OF-THE-ART METHODS ON THE KMU-FED DATASET

Methods	Accuracy
hierarchical WRF [27]	94.7%
CCNN [29]	97.3%
modified SqueezeNet 1.1 [30]	83.4%
DFER-GCViT (Our method)	98.2%

b) Results on FER2013 dataset: In the evaluation on the FER2013 dataset, our proposed DFER-GCViT method was compared against several state-of-the-art methods, including (1) a deep FER model integrating both CNNs and the ensemble deep learning algorithm [31], (2) an adapted VGGNet configuration consisting of four convolutional stages and three fully connected layers [32], and (3) a modified version of SqueezeNet 1.1 [30]. Remarkably, our DFER-GCViT achieved an accuracy of 73.78%, surpassing all the compared methods. Specifically, it outperformed deep FER by 1.87% and the modified SqueezeNet by an impressive 12.69%. Although our model achieved only slightly higher accuracy than the adapted VGGNet by 0.5%, the results were still interesting. Additionally, we conducted a similar comparison on the KMU-FED dataset with the original GCViT method. The results are summarized in Table III. Our proposed method exhibited superior performance, outperforming the original GCViT method by 1% (achieving an accuracy of 73.8%

compared to the original method's accuracy of 72.8%). These findings highlight the effectiveness of our proposed method in accurately recognizing facial expressions on the FER2013 dataset.

We observed that the accuracy on the FER2013 dataset is lower in comparison to the accuracy achieved on the KMU-FED dataset. This disparity can be attributed to the challenges present within the FER2013 dataset, such as the presence of corrupted images, noisy images, and low-resolution images.

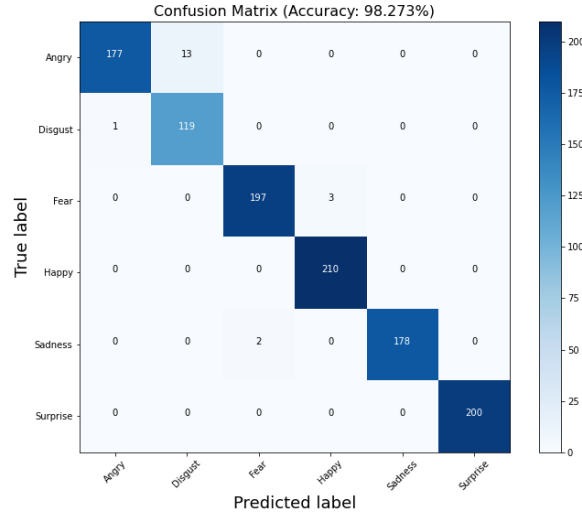
TABLE III
COMPARISON BETWEEN THE PROPOSED METHOD AND STATE-OF-THE-ART METHODS ON THE FER2013 DATASET

Methods	Accuracy
deep FER [31]	71.91%
VGG [32]	73.28%
modified SqueezeNet 1.1 [30]	61.09%
DFER-GCViT (Our method)	73.78%

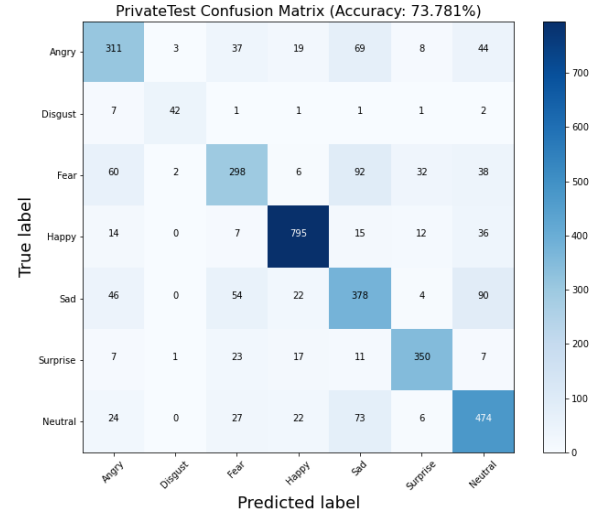
c) Analysis: In this subsection, we further analyze the results of expression recognition using the DFER-GCViT model on the two datasets: KMU-FED and FER2013. We first analyze the confusion matrices in Fig. 5, which offer valuable insights into the model's performance in recognizing various facial expressions. Subsequently, we examine the visualization heatmaps (see Fig. 6) to gain a further understanding of the model's decision-making process.

The confusion matrix in Fig. 5a, provides insights into the model's performance in recognizing different facial expressions on the KMU-FED dataset. The expressions of happiness and surprise achieve the highest accuracy, with no instances of misclassification. Conversely, the expressions of disgust, fear, and sadness exhibit a relatively small number of misclassifications. However, the model struggles to accurately distinguish between the anger and disgust expressions, resulting in a significant number of misclassifications for the anger. This difficulty can be attributed to the similarities in facial features and muscle movements associated with anger and disgust expressions. Moving on to the FER2013 dataset, as depicted in Fig. 5b, similar trends are observed. Once again, the expressions of happiness and surprise demonstrate the highest performance, with accurate classification. The model also appears confident in recognizing the angry and sad classes. However, similar to the observations from the KMU-FED dataset, these expressions may share visual similarities with other expressions, making it more challenging for the model to distinguish them accurately.

In order to gain insights into the model's reasoning and decision-making process, we utilized the Grad-CAM (Gradient-weighted Class Activation Mapping) [33] visualization technique to generate heatmaps for the DFER-GCViT model. Grad-CAM highlights the regions in the input image that are important for making the classification. We focused on visualizing the layers before the classification layer, specifically the layers in the block of stage 4 from the GC-ViT architecture (see Fig. 3), as these layers capture high-level



(a) KMU-FED



(b) FER2013

Fig. 5. Confusion matrices of the KMU-FED dataset and FER2013 dataset.



(a) KMU-FED

(b) FER2013

Fig. 6. Heatmap visualization of DFER-GCViT Using Grad-Cam technique for facial expressions on KMU-FED and FER2013 datasets.

features and semantic information. The colors in the heatmap represent the importance of different regions of the face. Warmer colors such as red and yellow indicate the most relevant regions during the classification process, while cooler colors like blue and green represent less important regions.

By analyzing the heatmaps, we can observe how the model localizes and focuses on specific regions of the face for different facial expressions. In the case of the KMU-FED dataset, as shown in Fig. 6a, the model exhibits concentration on discriminative features for the target expressions. Most expressions focus primarily on the mouth region, highlighting its important role in distinguishing between these facial expressions. Additionally, specific expressions show certain features that contribute more to the classification. However, for the anger class, we observe that the model focuses on the wrong region, neglecting other important features. This emphasizes the lower performance of the model in accurately classifying the anger expression, as indicated by the confusion matrix. Similarly, with the FER2013 dataset depicted in Fig.

6b, the model's focus varies for each specific expression, highlighting the regions such as eyes and eyebrows that contribute the most to the classification decision. Understanding these patterns provides valuable insights into the visual cues that drive the model's decision-making process and can aid in the interpretation and improvement of the facial expression recognition system.

V. CONCLUSIONS

In this paper, we have explored the application of GC-ViT in the facial expression recognition task, by introducing a novel transfer learning-based method utilizing the global context vision transformer DFER-GCViT to leverage its ability to capture both local and global features using a self-attention mechanism. Our approach focused on modifying the classifier head of the original GC-ViT by adding a new fully connected layer to accurately recognize facial expressions and emotions exhibited by drivers. To address the lack in the amount of data, commonly associated with vision transformers, we applied

data augmentation techniques to augment the training data in terms of both quantity and diversity and avoid overfitting. To evaluate the effectiveness of our proposed method, we conducted experiments on two distinct datasets: the KMUFED dataset, which simulates real driving scenarios and presents challenges that can arise in such contexts, and the FER2013 wild dataset, comprising images from unconstrained environments. Our proposed method exhibited competitive performance in facial expression recognition when compared to state-of-the-art methods. These results demonstrate the effectiveness of our approach in improving accuracy, allowing for accurate recognition of facial expressions despite the inherent challenges in real-world scenarios, such as occlusions, head pose variations, and varying illumination conditions. This research highlights the potential of vision transformers in the field of driver facial expression recognition, offering a novel perspective compared to previous methods, which helps the researchers to make further advancements in driver facial expression recognition methods, ultimately leading to improved driving safety.

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